



FOAM-WATER SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE

Date:

Inspector:

System:

Location:

General

System designation

Building

Location of control valve

Has system been modified since last inspection?

Yes

No

What is hazard protected?

Valves

How are valves supervised?

Sealed

Locked

Tamper switch

Are valves identified with signs?

Yes

No

Fire Department Connections

Location

Are identification signs provided?

Yes

No

Operating Instructions

Are operating instructions present?

Yes

No

Notes:





FOAM-WATER SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Property Name:

Inspector:

Property Address:

Contract No.:

Property Phone Number:

Date:

Inspection Frequency:

Daily

Weekly

Monthly

Quarterly

Annual

Five-Year

Inspections: Daily

Valve (Cold Weather/Heating Season Only)

Yes

No

N/A

Enclosure, not equipped with a low temperature alarm, is inspected during cold weather to verify a minimum temperature of 40°F (4°C)

Inspections: Weekly

Backflow

Yes

No

N/A

Isolation valves — open position and locked or supervised

Yes

No

N/A

RPA and RPDA — differential-sensing relief valve operating correctly

Control Valves

Yes

No

N/A

In the correct (open or closed) position

Yes

No

N/A

Sealed

Yes

No

N/A

Accessible

Yes

No

N/A

Post Indicator Valves (PIVs) are provided with correct wrenches

Yes

No

N/A

Free from damage or leaks

Yes

No

N/A

Proper signage

Deluge/Preaction Valve

Yes

No

N/A

Enclosure, where equipped with low temperature alarm, is inspected during cold weather to verify a minimum temperature of 40°F (4°C)

Master Pressure-Regulating Device

Yes

No

N/A

Downstream pressures are in accordance with design criteria psi

Yes

No

N/A

Supply pressure is in accordance with design criteria psi

Yes

No

N/A

Free of damage or leaks

Yes

No

N/A

Trim in good operating condition

Inspections: Monthly

Yes

No

N/A

Gauges are in good operating condition

Yes

No

N/A

Gauges — normal air or nitrogen pressure maintained when not supervised at a constantly attended location psi

Yes

No

N/A

Gauge on system side of dry valve reads proper ratio of air or nitrogen when not supervised at a constantly attended location psi

Yes

No

N/A

Gauge on quick-opening device reads the same as system side dry valve gauge when not supervised at a constantly attended location psi

NFPA 25 Handbook (p. 2 of 9)



FOAM-WATER SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Inspections: Monthly

Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A

Control Valves (locked or supervised)

In the correct (open or closed) position
Locked or supervised
Accessible
Post Indicator Valves (PIVs) are provided with correct wrenches
Free from damage or leaks
Proper signage

Deluge/Preaction Valve

Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A

Free from physical damage or leaks
Electrical components are in service
Valve seal not leaking
Trim valves are in the correct (open or closed) position

Proportioning System(s)

Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A

Foam concentrate quantity meets original design requirements
Strainers inspected
Pressure vacuum vent is operating freely
Gauges are in good operating condition
Sensing line valves are open
Foam liquid pump provided with power
Free of corrosion
Automatic drain valves are free and opened
Valves in correct position

Nozzles

Yes	No	N/A
Yes	No	N/A
Yes	No	N/A

In place, aimed, and pointed in the direction intended
Free from external loading and corrosion
Caps or plugs are in place (where required)

Inspections: Quarterly

Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A

Gauges — normal air or nitrogen pressure maintained when supervised at a constantly attended location psi
Gauge on system side of dry valve reads proper ratio of air or nitrogen when supervised at a constantly attended location psi
Gauge on quick-opening device reads the same as system side dry valve gauge when supervised at a constantly attended location psi
Gauge on supply side of valve reads normal psi
Alarm devices — free of damage
Hydraulic design information sign is securely attached to riser and legible

Alarm Valves/Riser Check

Yes	No	N/A
Yes	No	N/A
Yes	No	N/A
Yes	No	N/A

In appropriate open or closed position
Free of damage
Accessible
Retard chamber/alarm drains not leaking

FOAM-WATER SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Inspections: Quarterly

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Inspections: Annual

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Yes No N/A

Pressure-Reducing Valve

In the open position and not leaking

Maintaining downstream pressure

In good condition, with handwheels installed and unbroken

Foam Concentrate Strainer(s)

Blow-down valve closed and plugged

Drainage

Method — in good operating condition

Retention embankments or dikes are in good condition

Control Valves (Electronically Supervised)

In the correct (open or closed) position

Electronically supervised

Accessible

Post Indicator Valves (PIVs) are provided with correct wrenches

Free from damage or leaks

Proper signage

Low temperature alarm free of damage

Support/Hangers

In good condition (no missing or damaged paint or coating, rust, or corrosion)

No damaged or missing hangers bracing and support

Securely attached to structure and piping

Pipes and Fittings

In good condition and no external corrosion

No leaks or mechanical damage

Correct alignment — no external loads

Low point drains maintained and in proper working order

Rubber-gasketed fittings in good condition

Sprinklers (visible)

No damage or leaks

Free of corrosion, foreign material, or paint

Fluid in glass bulbs

Spare sprinklers — proper number and type, including installation wrench

Loading — sprinklers are free of dust

Escutcheons/cover plates are present and installed correctly

Clearance — minimum clearance between sprinklers and storage

FOAM-WATER SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Inspections: Annual

Yes	No	N/A
-----	----	-----

Bladder Tank

Presence of foam in water surrounding bladder

Deluge Valve

Yes	No	N/A
-----	----	-----

Interior is in good working condition

Yes	No	N/A
-----	----	-----

Detection device in good condition

Inspections: Five Years

Yes	No	N/A
-----	----	-----

Strainers — per manufacturer's instructions

Yes	No	N/A
-----	----	-----

Alarm valve interior in good condition

Yes	No	N/A
-----	----	-----

Check valve — internal moves freely and in good condition

Yes	No	N/A
-----	----	-----

Obstruction inspection — no foreign or obstructing material found

Yes	No	N/A
-----	----	-----

Backflow — internal inspection

Yes	No	N/A
-----	----	-----

Internal inspection of deluge valve strainers, filters, restricted orifices, and diaphragm chambers

Test: Quarterly

Yes	No	N/A
-----	----	-----

Alarm devices — water motor gong

Yes	No	N/A
-----	----	-----

Priming water level tested

Yes	No	N/A
-----	----	-----

Low air alarm tested per manufacturer's instructions

Yes	No	N/A
-----	----	-----

Main drain tested if sole supply is through a backflow preventer or pressure-reducing valve

Yes	No	N/A
-----	----	-----

Static psi Residual psi

Yes	No	N/A
-----	----	-----

Do main drain test results differ more than 10% from previous test?

Master Pressure-Regulating Device

Yes	No	N/A
-----	----	-----

Partial flow test performed to exercise valve

Test: Semiannual

Yes	No	N/A
-----	----	-----

Valve supervisory switch(es) function

Yes	No	N/A
-----	----	-----

Alarm devices (vane and pressure switch type) tested and functioning correctly

Test: Annual

Yes	No	N/A
-----	----	-----

All control valves operated through full range of motion and returned to normal position

Yes	No	N/A
-----	----	-----

Valve status test performed

Yes	No	N/A
-----	----	-----

Supervisory switch(es) function

Yes	No	N/A
-----	----	-----

Low temp alarm (if installed) at the beginning of heating season

Main Drain Test

Static psi Residual psi

Yes	No	N/A
-----	----	-----

Do main drain test results differ more than 10% from previous test?

NFPA 25 Handbook (p. 5 of 9)

FOAM-WATER SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Test: Annual

Backflow Preventer

Yes	No	N/A	Forward flow test at a minimum flow rate of the system demand
-----	----	-----	---

Master Pressure-Regulating Device

Yes	No	N/A	Full flow test compared to previous test results
-----	----	-----	--

Full Flow Test (Deluge Valve)

Yes	No	N/A	Unobstructed discharge from all nozzles/position correctly
-----	----	-----	--

Yes	No	N/A	Pressure reading at deluge valve psi
-----	----	-----	---

Yes	No	N/A	Pressure reading at most remote nozzle of sprinkler psi
-----	----	-----	--

Yes	No	N/A	Air maintenance functions correctly (if provided)
-----	----	-----	---

Preaction Partial Trip Test

Water pressure	(psi)	Air pressure	(psi)	
----------------	-------	--------------	-------	--

Tripping air pressure	(psi)	Trip time	(sec)	
-----------------------	-------	-----------	-------	--

Water delivery time	(min.)		(sec)	
---------------------	--------	--	-------	--

Foam Concentrate Strainer(s)

Yes	No	N/A	In proper working condition
-----	----	-----	-----------------------------

Operational Test

Yes	No	N/A	Foam-water system(s) responded as designed
-----	----	-----	--

Discharge time	(min.)		(sec)	
----------------	--------	--	-------	--

Yes	No	N/A	Manual actuation device functions as intended
-----	----	-----	---

Concentration Test

Yes	No	N/A	Foam sample measured
-----	----	-----	----------------------

Yes	No	N/A	Foam concentration within 10% of acceptance test results
-----	----	-----	--

Yes	No	N/A	Foam concentrate not more than 10% below minimum design standard
-----	----	-----	--

Test: Three Years

Preaction Valve Trip Test Open Control Valve

Water pressure	(psi)	Air pressure	(psi)	
----------------	-------	--------------	-------	--

Tripping air pressure	(psi)	Trip time	(sec)	
-----------------------	-------	-----------	-------	--

Water delivery time	(min.)			
---------------------	--------	--	--	--

Yes	No	N/A	Results compared to previous results
-----	----	-----	--------------------------------------

Yes	No	N/A	Preaction system tested for air leakage
-----	----	-----	---

FOAM-WATER SPRINKLER SYSTEMS INSPECTION, TESTING, AND MAINTENANCE (Continued)

Test: Five Years

Yes	No	N/A	
			Gauges tested or replaced

Routine Maintenance

Yes	No	N/A	OS&Y — stems lubricated annually
-----	----	-----	----------------------------------

Yes	No	N/A	Proportioned ball drip disassembled, cleaned, and reassembled every five years. Date:
-----	----	-----	---

Yes	No	N/A	Foam concentrate tank drained and flushed every ten years. Date:
-----	----	-----	--

Yes	No	N/A	Foam concentrate tank inspected internally and externally for corrosion every ten years. Date:
-----	----	-----	--

Yes	No	N/A	Foam concentrate tank hydrostatically tested every ten years. Date:
-----	----	-----	---

Yes	No	N/A	Bladder tank proportioner sight glass cleaned every ten years. Date:
-----	----	-----	--

Yes	No	N/A	Balancing valve diaphragm flushed every five years. Date:
-----	----	-----	---

Yes	No	N/A	Pressure vacuum vents serviced every five years. Date:
-----	----	-----	--

Comments

Signature:

Date:

Contractor Name:

Contractor Address:

License/Certification No.: